

SOC TASK 2 – INCIDENT RESPONSE REPORT

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Tools Used: Splunk Enterprise (local), Sample SOC Logs

1. Executive Summary

As part of the SOC internship project, simulated security logs were ingested into Splunk SIEM to perform real-time alert monitoring and basic incident response.

During analysis, multiple suspicious events were identified:

- **Repeated failed login attempts** from foreign IPs
- **A malware detection alert**
- **Unusual successful login from a risky IP**

These indicate potential brute force activity, credential misuse, and a malware infection. This report documents findings, event timelines, severity classification, and recommended remediation actions.

2. Scope

The analysis covers the following log types from the sample dataset:

- Authentication logs
- Network connection attempts
- Malware detection logs
- User activity logs

Objective:

1. Identify suspicious events
2. Classify incidents based on severity
3. Document findings & mitigation steps
4. Produce dashboard visualizations in Splunk

3. Data Ingestion Summary

- SIEM Tool: **Splunk Enterprise (local instance)**
- Source File: SOC_Task2_Sample_Logs.txt
- Sourcetype: syslog
- Index: soc_task2
- Events successfully parsed: **50+**
- Fields extracted: user, ip, action, threat, etc.

4. Summary of Suspicious Events Identified

✓ 4.1 Brute Force Login Attempts (High Severity)

Several users had:

- multiple failed login attempts
- occurring in short intervals
- from foreign/risky IP addresses (e.g., **203.0.113.77**, **172.16.0.x**)

This behavior indicates password-guessing or credential abuse.

✓ 4.2 Malware Detection Alert (High Severity)

One event indicated:

- **action = malware detected**
- **threat = Trojan Detected**
- Source user: bob
- IP: 10.0.0.5

This is a confirmed malware event and requires immediate isolation of the host.

✓ **4.3 Suspicious Successful Login (Medium Severity)**

A successful login was observed:

- User: bob
- IP: 172.16.0.3
- Behavior inconsistent with user's normal IP pattern

This may indicate credential compromise.

✓ **4.4 Repeated Connection Attempts (Medium Severity)**

Certain users had repeated:

- action=connection attempt
- from changing IP addresses

This may indicate network scanning or recon activity.

✓ **4.5 Low-Risk Normal Failures (Low Severity)**

Single failed login attempts without repetition — considered normal end-user behavior.

5. Incident Classification Table

Alert ID	Event	User	IP	Severity	Reason
A1	Multiple failed logins	david/alice	203.0.113.77	High	Possible brute force
A2	Malware Detected	bob	10.0.0.5	High	Trojan alert
A3	Unusual Login Success	bob	172.16.0.3	Medium	Possible credential misuse
A4	Repeated connection attempts	charlie	198.51.100.42	Medium	Potential recon
A5	Single login failure	Various	Various	Low	Normal user error

6. Event Timeline Summary

Time	User	IP	Action	Notes
03/07/2025 09:02	david	203.0.113.77	login failed	Multiple failures
03/07/2025 07:02	alice	203.0.113.77	login failed	Pattern resembles brute force
03/07/2025 05:48	bob	10.0.0.5	malware detected	Trojan
03/07/2025 05:18	bob	172.16.0.3	login success	Suspicious IP
03/07/2025 04:23	charlie	198.51.100.42	login failed	Possible recon

7. Technical Analysis (Per Incident)

Incident 1 — Brute Force Attempts

Evidence:

- Multiple failed login attempts from IP 203.0.113.77
- Targeted multiple users
- Pattern: high-frequency login failures

Risk: Possible credential-stuffing or brute force attack

Recommendation: Enable rate-limiting, enforce MFA

Incident 2 — Malware Detection

Evidence:

- action=malware detected
- threat=Trojan Detected
- Affected user: bob
- Host: 10.0.0.5

Risk: High — Trojan infection on internal workstation

Recommendation:

- Isolate host immediately
- Run antivirus / EDR scan
- Reset access tokens/passwords

Incident 3 — Suspicious Login Success

Evidence:

- Successful login from IP outside typical range
- Occurred shortly after failed attempts

Risk: Medium

Recommendation:

- Force password reset
- Review session logs

Incident 4 — Connection Attempts

Evidence:

- Repeated connection attempts from various subnets
- Potential scanning behavior

Risk: Medium

Recommendation: Monitor IP, block if persistent

8. Recommended Remediation Steps

Containment

- Lock or monitor affected user accounts
- Isolate the malware-infected host
- Block suspicious IPs temporarily

Eradication

- Perform full malware removal procedure
- Patch vulnerable systems
- Enforce password resets

Recovery

- Restore system integrity
- Monitor for recurring attacks
- Implement MFA and strong password policy

Long-Term Prevention

- Enable brute force detection alerts
- Deploy EDR solution
- Regular SOC monitoring
- Regular user security training

9. Conclusion

The Splunk analysis of SOC_Task2 logs revealed legitimate indicators of brute force activity, malware infection, and abnormal user authentication patterns. Quick mitigation steps such as password resets, host isolation, and malware scanning can reduce the threat significantly.

This exercise demonstrates fundamental SOC skills:

- Log analysis
- Alert triage
- Severity classification
- Incident documentation
- Dashboard creation

10. Appendix

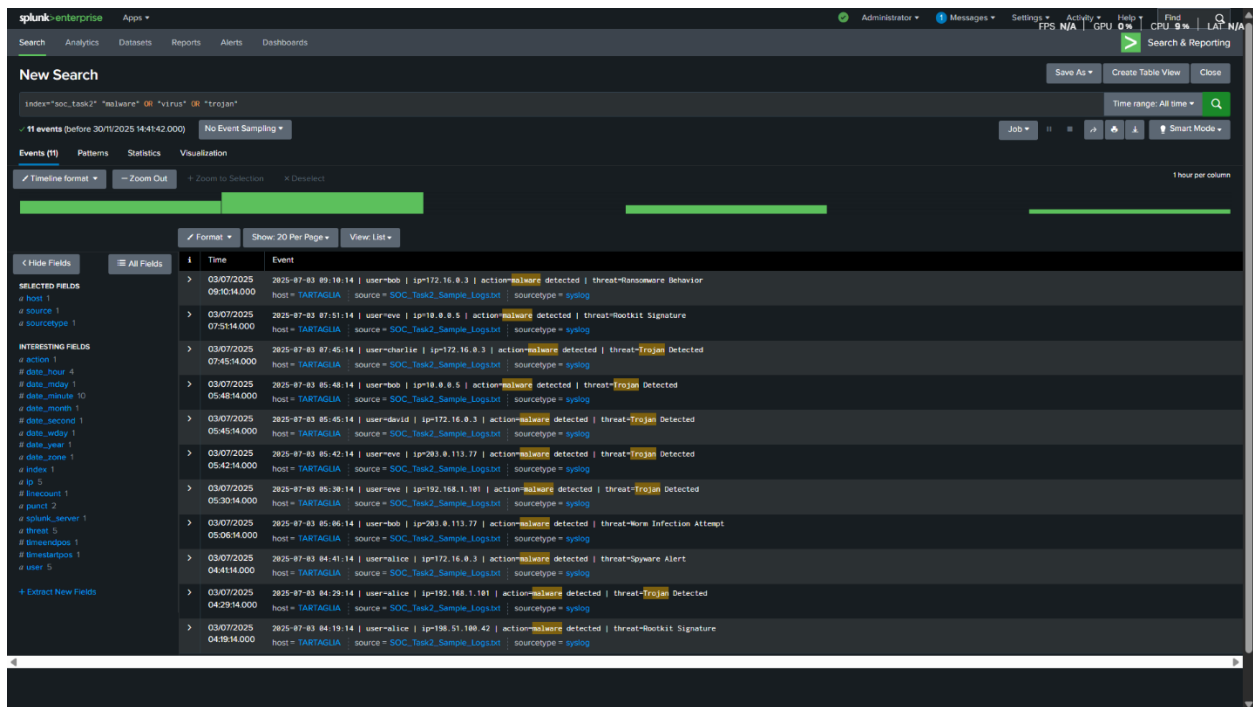
a. Brute Force Login

The screenshot shows the Splunk search interface with the following details:

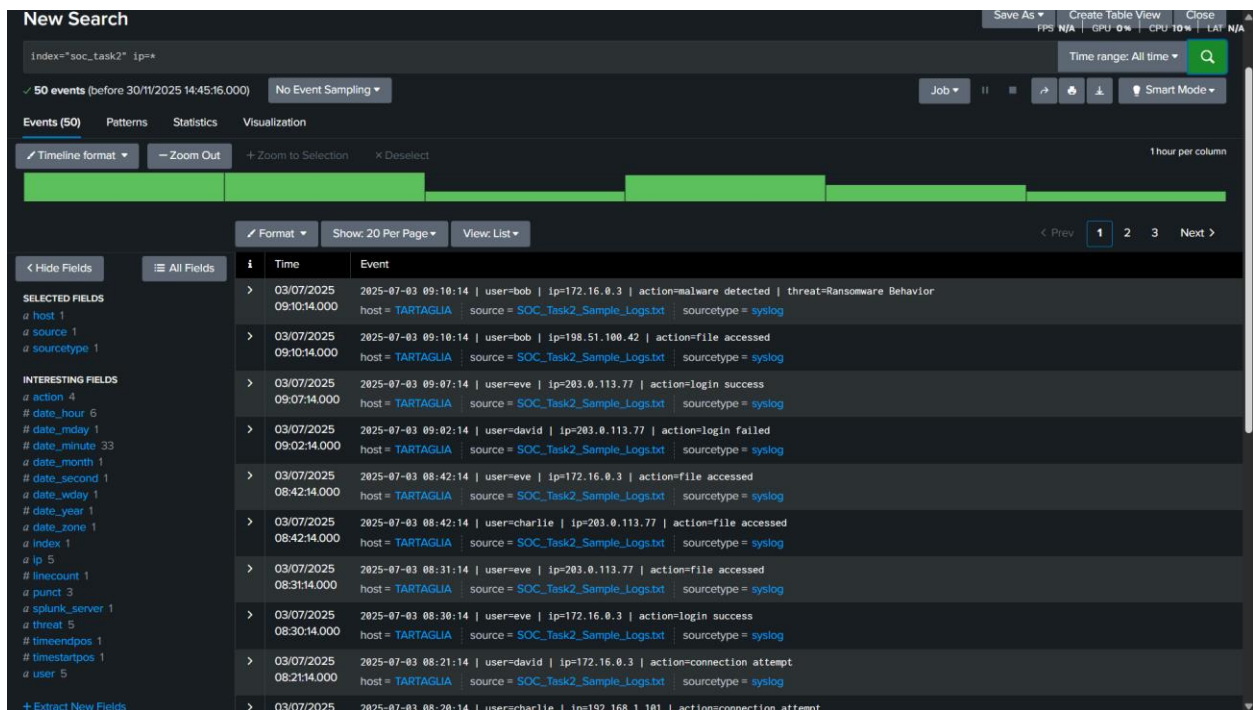
- Search Query:** `index="soc_task2" "failed" OR "authentication failure"`
- Time Range:** All time
- Results:** 5 events (before 30/11/2025 14:39:59.000)
- Fields:** host, source, sourcetype
- Events:** 5 events (before 30/11/2025 14:39:59.000)
- Fields:** host, source, sourcetype
- Fields:** host, source, sourcetype

Time	Event
03/07/2025 09:02:14.000	2025-07-03 09:02:14 user=david ip=203.0.113.77 action=login failed host = TARTAGLIA source = SOC_Task2_Sample_Logs.txt sourcetype = syslog
03/07/2025 07:02:14.000	2025-07-03 07:02:14 user=alice ip=203.0.113.77 action=login failed host = TARTAGLIA source = SOC_Task2_Sample_Logs.txt sourcetype = syslog
03/07/2025 04:47:14.000	2025-07-03 04:47:14 user=bob ip=10.0.0.5 action=login failed host = TARTAGLIA source = SOC_Task2_Sample_Logs.txt sourcetype = syslog
03/07/2025 04:23:14.000	2025-07-03 04:23:14 user=bob ip=172.16.0.3 action=login failed host = TARTAGLIA source = SOC_Task2_Sample_Logs.txt sourcetype = syslog
03/07/2025 04:23:14.000	2025-07-03 04:23:14 user=charlie ip=198.51.100.42 action=login failed host = TARTAGLIA source = SOC_Task2_Sample_Logs.txt sourcetype = syslog

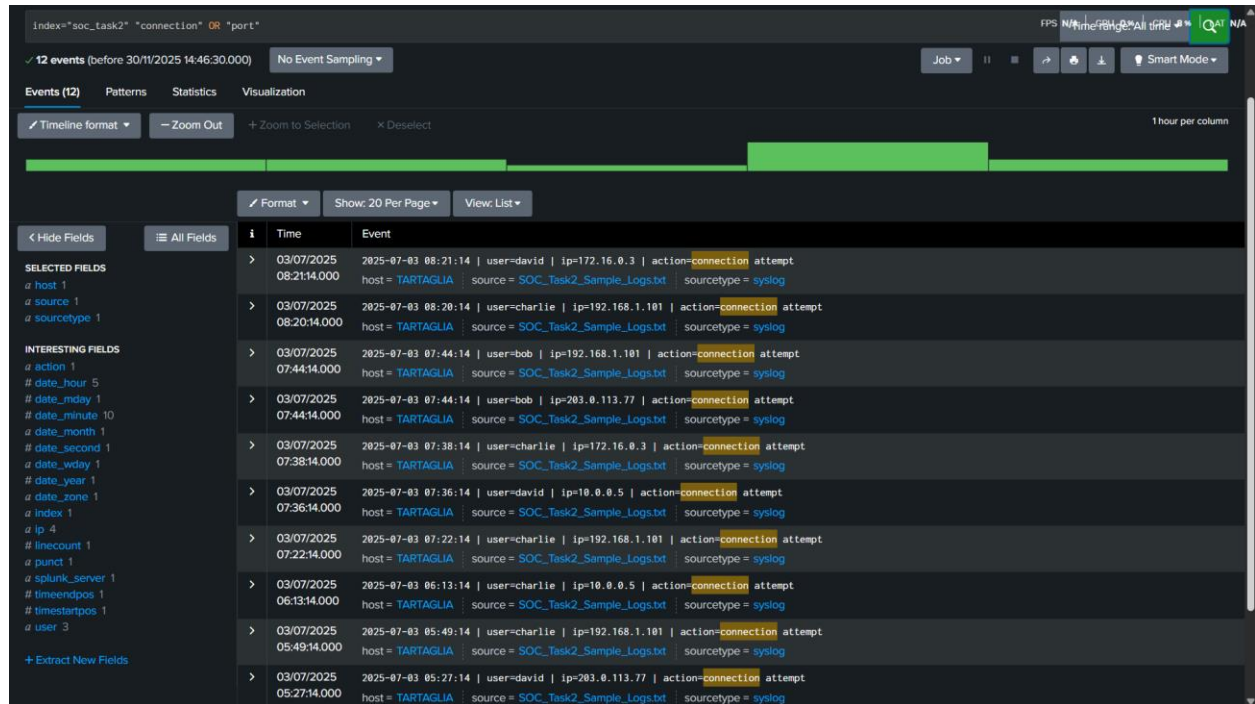
b. Malware Detection



c. Suspicious Successful Logins From IP



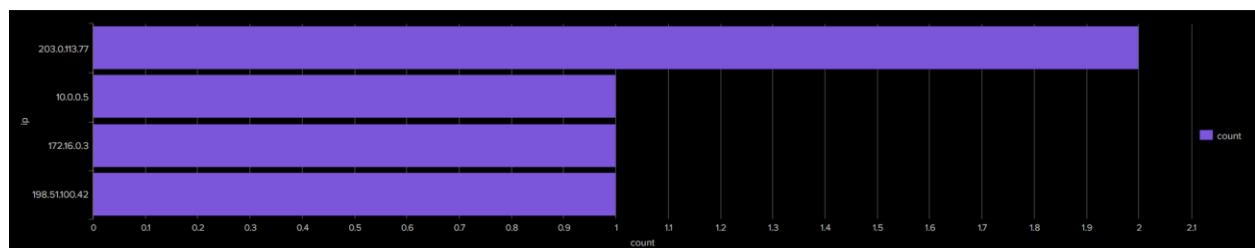
d. Connection Attempts



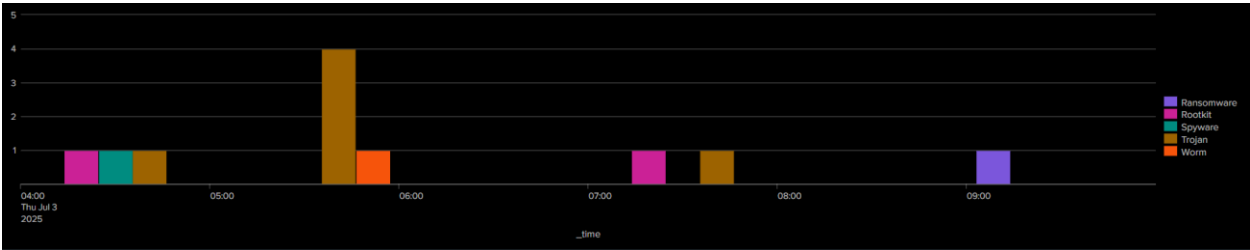
Line chart of failed logins over time



Bar chart of top IPs



Malware alert visualization



Complete Dashboard View

